and an exogeneous continuous covariate Z_{ij} is generated from Normal$(0,1)$. The binary treatment $A_{ij}, j \in [J]$ is generated from Bernoulli(0.4). In the time-varying functions we scaled both time indices to take values in $[0,1]$: let $\tilde{j} = j/J$ and $\tilde{t} = t/T$, and we generated the proximal outcome as

$$\begin{aligned} Y_{ijt} = A_{ij}\big\{\beta_0 + \beta_1(\tilde{j}) + \beta_{21}(\tilde{t}) + S_{ij}\beta_{22}(\tilde{t})\big\} \\ + \alpha_0 + \alpha_{11}(\tilde{j}) + S_{ij}\alpha_{12}(\tilde{j}) + \alpha_{21}(\tilde{t}) + Z_{ij}\alpha_{22}(\tilde{t}) + e_{ijt}, \end{aligned}$$

where the e_{ijt}'s are marginally Normal$(0,1)$, independent across different i or j, and for the i-th individual at the j-th decision point $\mathrm{Corr}(e_{ijt}, e_{ijs}) = 0.4$ for $t \neq s$. The causal excursion effect is thus

$$\mathrm{CEE}(j, t, S_{ij}) = \beta_0 + \beta_1(\tilde{j}) + \beta_{21}(\tilde{t}) + S_{ij}\beta_{22}(\tilde{t}). \tag{8}$$

We set $\alpha_0 = 0.35$ and $\beta_0 = 0.7$. For the time-varying functions, we constructed two relatively complex functions $h_1(\cdot), h_2(\cdot)$ and two relatively simple functions $h_3(\cdot), h_4(\cdot)$, all defined on $[0,1]$ and adapted from Wahba (1983):

$$h_1(\cdot) = \frac{6}{10}g_{30,17}(\cdot) + \frac{4}{10}g_{3,11}(\cdot), \qquad h_2(\cdot) = \frac{1}{3}g_{20,5}(\cdot) + \frac{1}{3}g_{12,12}(\cdot) + \frac{1}{3}g_{7,30}(\cdot)$$
$$h_3(\cdot) = \frac{1}{3}g_{10,5}(\cdot) + \frac{1}{3}g_{7,7}(\cdot) + \frac{1}{3}g_{5,10}(\cdot), \qquad h_4(\cdot) = \frac{1}{3}g_{8,5}(\cdot) + \frac{1}{3}g_{7,7}(\cdot) + \frac{1}{3}g_{5,8}(\cdot).$$

where $g_{a,b}(\cdot)$ denotes the probability density function of a Beta distribution with shape parameters a, b (Figure 2). We considered the following simulation settings :

- (Truth being complex functions.) $\beta_1 = h_1$, $\beta_{21} = h_1$, $\beta_{22} = h_2$, $\alpha_{11} = h_1$, $\alpha_{12} = h_3$, $\alpha_{21} = h_1$, $\alpha_{22} = h_2$.

- (Truth being simple functions.) $\beta_1 = h_3$, $\beta_{21} = h_3$, $\beta_{22} = h_4$, $\alpha_{11} = h_3$, $\alpha_{12} = h_3$, $\alpha_{21} = h_3$, $\alpha_{22} = h_2$.

In the first stage of the estimator, we used the working model for expected outcome under

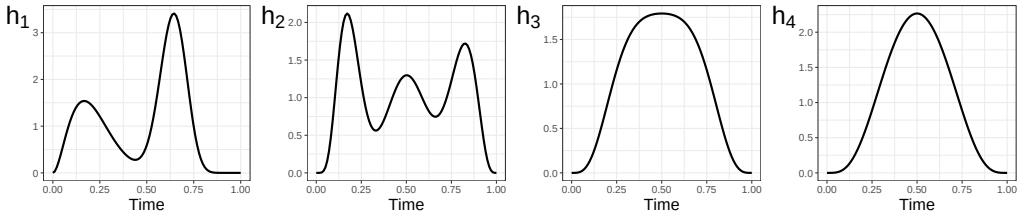

Figure 2: Functions h_1, h_2, h_3, and h_4 used in the simulations.

Table 1: MSE and coverage probability of the estimated intercept in CEE ($\hat{\beta}_0$) under complex (top half) and simple (bottom half) settings in the simulation.

Simulation setting	Sample size	MSE	Coverage Probability
	n = 20	0.061	0.911
Complex functions	n = 50	0.017	0.915
	n = 100	0.007	0.938
	n = 20	0.057	0.911
Simple functions	n = 50	0.016	0.913
	n = 100	0.007	0.938

no treatment: $\eta(j, t, H_{ij}) = \alpha_0 + \alpha_1(\tilde{j}) + \alpha_{21}(\tilde{t}) + S_{ij}\alpha_{22}(\tilde{t})$. We misspecified the working model by omitting $Z_{ij}\alpha_{22}(\tilde{t})$ to assess the robustness of the estimator. The causal excursion effect is correctly modeled as (8). We used cubic splines with evenly spaced knots between $[0, 1]$ as the nonparametric function approximators for $\beta_1(\cdot)$, $\beta_{21}(\cdot)$, $\beta_{22}(\cdot)$, $\alpha_1(\cdot)$, $\alpha_{21}(\cdot)$, and $\alpha_{22}(\cdot)$. The numbers of knots were selected for each simulated data set using 10-fold cross-validation and optimized using a one-pass coordinate search for computational efficiency.

We considered sample sizes $n = 20, 50, 100$ and conducted each simulation for 1000 replicates. For the setting with complex time-varying functions, as the sample size increases,

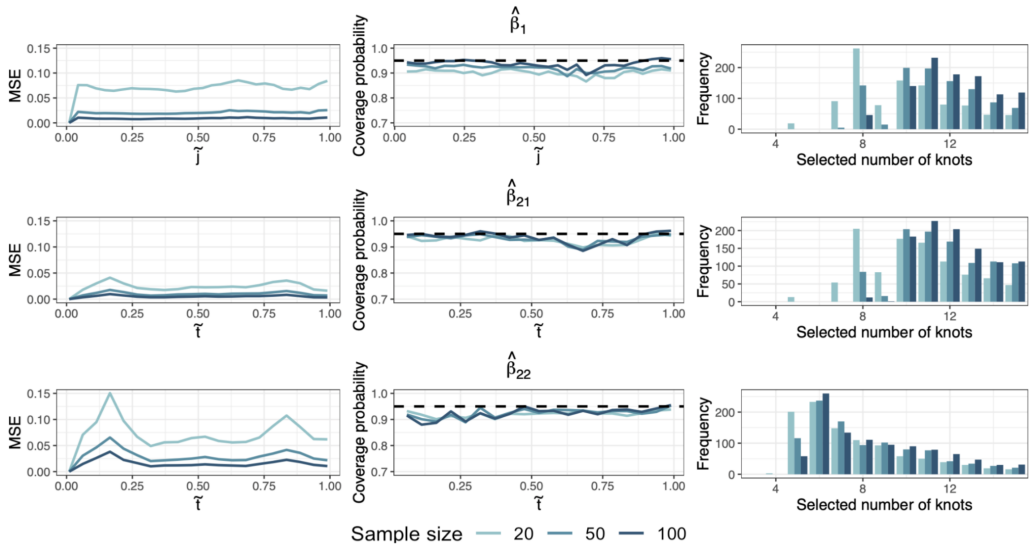

Figure 3: Mean squared error (left), coverage probability of 95% confidence interval (center), and selected number of knots (right) for $\hat{\beta}_1(\tilde{j})$, $\hat{\beta}_{21}(\tilde{t})$, and $\hat{\beta}_{22}(\tilde{t})$ under the complex function setting in the simulation.

the mean squared error (MSE) of the estimated intercept in CEE ($\hat{\beta}_0$) decreases (Table 1, top half), and the point-wise MSE of $\hat{\beta}_1(\tilde{j})$, $\hat{\beta}_{21}(\tilde{t})$, and $\hat{\beta}_{22}(\tilde{t})$ decreases (Figure 3, left column). This verifies the consistency result (Theorem 1). In addition, the coverage probability of 95% confidence interval is close to the nominal level even with a small sample size such as $n = 20$ (Figure 3, middle column). Lastly, as the sample size increases, the number of selected knots increases slightly (Figure 3, right column).

For the setting with simple time-varying functions, similar observations can be made regarding the decreasing MSE and close-to-nominal coverage probability (Table 1, bottom half; Figure 4). In addition, comparing the right column between Figures 3 and 4 shows that the selected number of knots is larger when the time-varying functions are complex.